AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Centersquare IAD1-B, VA -- station KIAD, America/New_York
Committed pair	OSM 701930919 -> 52094270 CyrusOne NVA8 / Centersquare IAD1-A
Facade gap	84.8 m between the two halls
Weather record	43,763 real hourly records from KIAD
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.5421 C at 210 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-09-09 (chatter)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 1 of 24 hours with 1 mode change. 1 hour was safe but still ran chillers, because the schedule could not afford them.

Agent	1 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	1 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s) 1 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	refusal	21.348	21.0	21.110	1.094
01	mechanical	no	refusal	21.046	21.0	20.560	1.007
02	mechanical	no	refusal	21.593	21.0	21.110	1.173
03	mechanical	no	dew point	21.203	21.0	20.560	0.968
04	mechanical	no	dew point	20.478	21.0	19.440	0.993
05	mechanical	no	refusal	20.778	21.0	18.890	1.216
06	mechanical	no	dew point	20.445	21.0	18.890	0.946
07	mechanical	no	dew point	21.484	21.0	19.440	1.538

08	mechanical	no	dew point	23.640	21.0	21.117	2.095
09	mechanical	no	refusal	24.436	21.0	20.560	2.421
10	mechanical	no	dew point	24.603	21.0	20.560	2.225
11	mechanical	no	dry-bulb	25.136	21.0	21.670	1.740
12	mechanical	no	dew point	24.555	21.0	22.780	1.278
13	mechanical	no	refusal	24.383	21.0	23.330	1.206
14	mechanical	no	refusal	25.536	21.0	25.000	1.350
15	mechanical	no	refusal	25.685	21.0	25.000	1.171
16	mechanical	no	refusal	25.752	21.0	26.243	1.141
17	mechanical	no	refusal	26.366	21.0	26.110	1.477
18	mechanical	no	dry-bulb	24.368	21.0	23.330	1.339
19	mechanical	no	dry-bulb	23.493	21.0	21.110	1.383
20	mechanical	no	dry-bulb	22.907	21.0	20.000	1.495
21	mechanical	yes	-	20.680	21.0	17.780	1.652
22	mechanical	no	refusal	20.213	21.0	18.890	1.402
23	FREE	yes	-	18.547	21.0	16.670	1.165

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

01:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

02:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

03:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 21.203 C would have passed. The outside air is too HUMID: the dew-point bound is 19.72 C against a 15.0 C maximum, failing by 4.72 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

04:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 20.478 C would have passed. The outside air is too HUMID: the dew-point bound is 18.61 C against a 15.0 C maximum, failing by 3.61 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

05:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

06:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 20.446 C would have passed. The outside air is too HUMID: the dew-point bound is 18.34 C against a 15.0 C maximum, failing by 3.34 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

07:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 21.484 C would have passed. The outside air is too HUMID: the dew-point bound is 17.77 C against a 15.0 C maximum, failing by 2.77 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

08:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 23.640 C would have passed. The outside air is too HUMID: the dew-point bound is 18.06 C against a 15.0 C maximum, failing by 3.06 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

09:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

10:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 24.603 C would have passed. The outside air is too HUMID: the dew-point bound is 18.89 C against a 15.0 C maximum, failing by 3.89 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.136 C against a 21.0 C limit, so it fails by 4.136 C. It would take a limit 4.136 C higher, or a bound 4.136 C tighter, to change this hour.

12:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 24.555 C would have passed. The outside air is too HUMID: the dew-point bound is 18.89 C against a 15.0 C maximum, failing by 3.89 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

13:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

14:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

15:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

16:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

17:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.368 C against a 21.0 C limit, so it fails by 3.368 C. It would take a limit 3.368 C higher, or a bound 3.368 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.493 C against a 21.0 C limit, so it fails by 2.493 C. It would take a limit 2.493 C higher, or a bound 2.493 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.908 C against a 21.0 C limit, so it fails by 1.908 C. It would take a limit 1.908 C higher, or a bound 1.908 C tighter, to change this hour.

21:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.680 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.547 C, which is 2.453 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.166 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.2123 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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